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Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circumf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																		2 He 4.003
2	3 Li 6.941	4 Be 9.012												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305												13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956		22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906		40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	*	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	**	104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

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Day 2. Grade 10th. Questions 11, 12 and 13.



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Question 11. Miscellanea are back (15% of total; only 10th grade) Day 2

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	Total
2	2	2	2	2	2	1	1	1	15

The professors of the National Chemistry Pre-selection have decided to test the theoretical knowledge of the new students and have made the following list of experimental facts, which they must explain.

11.1. Luis Guerra: Given the boiling temperatures of hydrogen halides: HF 20 °C, HCl –85 °C, HBr –66 °C and HI –35 °C. Why does HF have an unusually high melting point, whereas the rest of the elements vary in this property in an orderly manner?

11.2. Rolando: In period 3, the melting temperatures of the elemental substances first increase and then decrease: Na 98 °C, Mg 650 °C, Al 660 °C, Si 1414 °C, P 45 °C, S 119 °C, Cl –102 °C and Ar –189 °C.

11.3. Landrove: Graphite and graphene conduct the electric current, while diamond is an insulator, even though all these allotropes have the same composition.

11.4. Gerardo: The manganese oxides corresponding to oxidation states II and III are basic and reductant, IV is amphoteric and very stable, and VII is very acidic and strongly oxidizing.

11.5. Rafael: The stability of coordination compounds of the halides depends strongly on the central atom. For example, for iron(III), $[\text{FeF}_6]^{3-} \gg [\text{FeCl}_4]^- > [\text{FeBr}_4]^-$, while for mercury(II): $[\text{HgCl}_4]^{2-} < [\text{HgBr}_4]^{2-} \ll [\text{HgI}_4]^{2-}$. The Fe(III)-I⁻ or Hg(II)-F⁻ complexes are not known.

11.6. Luis Manuel: Noble gases form very few chemical compounds; however, oxides, fluorides and oxyfluorides of Xe are known.

11.7. Julieta: When the salts potassium iodide and lead(II) nitrate, both white, come into contact, no apparent change occurs. However, adding a few drops of water immediately causes an intense yellow coloration.

11.8. Eloy: The chemical elements Ga and Al have similar atomic radii, although they are in the same group 13 of the periodic table.

11.9. Rey Michel: The elements boron and silicon are similar in several properties. Both form a covalent atomic crystal lattice; they react with oxygen to form oxides in higher oxidation states and at very high melting temperatures; and they react with halogens to give molecular substances that are strongly hydrolyzed in water, yielding very acidic solutions.

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Day 2. Grade 10th. Questions 11, 12 and 13.



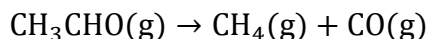
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Question 12. Acetaldehyde decomposition (15% of total; only 10th grade) Day 2

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	Total	Total
8	8	12	4	6	6	8	8	60	15

The thermal decomposition of acetaldehyde (ethanal) occurs at high temperatures, according to the equation.



The reaction at a certain temperature T was studied, obtaining the following results ($M \equiv \text{mol/L}$).

Experiment	$[\text{CH}_3\text{CHO}]_0$ (M)	Initial rate (M/s)
1	0,10	$1,1 \cdot 10^{-4}$
2	0,20	$4,4 \cdot 10^{-4}$
3	0,30	$9,9 \cdot 10^{-4}$

12.1. Determine the reaction order, the rate law, and the rate constant k_r with the appropriate units.

In other experiments, the dependence of the reaction rate constant, k_r , on temperature, T, in the range 700 to 1000 K was studied. The analysis of the data allowed us to adjust to a straight line when plotting $\ln(k_r)$ versus $(10^3 \cdot \frac{1}{T})$ with slope $-22,5$.

12.2. Calculate the value of the activation energy (in kJ/mol). Assume Arrhenius-like behavior.

The thermal decomposition of acetaldehyde proceeds via homolytic cleavage of a bond, generating radicals. One of the proposed mechanisms gives dihydrogen and ethane as minority by-products.

	Reaction	E_A (kJ/mol)
Initiation	$\text{CH}_3\text{CHO} \xrightarrow{k_1} \cdot \text{CH}_3 + \cdot \text{CHO}$	358
Propagation	$\text{CH}_3\text{CHO} + \cdot \text{CH}_3 \xrightarrow{k_2} \text{CH}_4 + \text{CH}_3\text{CO} \cdot$	8
	$\text{CH}_3\text{CO} \cdot \xrightarrow{k_3} \cdot \text{CH}_3 + \text{CO}$	59
	$\cdot \text{CHO} \xrightarrow{k_4} \cdot \text{H} + \text{CO}$	65
	$\cdot \text{H} + \text{CH}_3\text{CHO} \xrightarrow{k_5} \text{H}_2 + \text{CH}_3\text{CO} \cdot$	15
Termination	$2 \cdot \text{CH}_3 \xrightarrow{k_6} \text{C}_2\text{H}_6$	0

12.3. Use the steady-state approximation and find mathematical expressions for the variations in the concentrations of the radicals CHO , H , CH_3 and CH_3CO over time ($d[X]/dt$).

12.4. Why do you think the activation energy of the termination step is zero?

From the above general mechanism, it is inferred that the decomposition of acetaldehyde occurs by two simpler paths in parallel, with a total of equations.

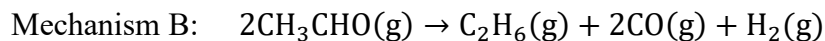
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**12.5.** Assign which steps in the general mechanism correspond to each of the simple mechanisms A and B.**12.6.** Taking into account your knowledge of thermodynamics, answer: a) How much is the enthalpy of formation of a simple substance in its standard state? b) What does the standard state symbol ($^\circ$) mean in enthalpy values? c) What is the meaning of molar caloric capacity?**12.7.** Calculate the enthalpy variations for the total reactions of mechanisms A and B at 298 K from the enthalpy variations in the table.

Substance	CH_3CHO	C_2H_6	CH_4	CO	H_2
ΔH_f° [kJ/mol]	-190	-84	-75	-110	?

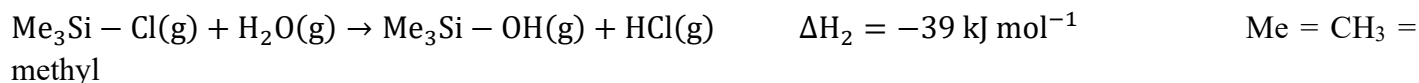
12.8. Represent the Lewis structures of H_2 , CO , CH_4 , C_2H_6 and CH_3CHO .**Question 13. Thermochemistry of silicon (20% of total; only 10th and 12th grades) Day 2**

To the memory of Professor Dr. Jacques Rieumont Briones.

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	Total	Total
2	18	6	14	12	18	4	2	4	80	20

Silicon shares characteristics with carbon, such as being in the same family 14, forming long chains (with oxygen in silicones), forming zeolites with catalytic capacity, and having four valence states. This gives it the hypothetical ability to support life forms outside the Earth.

To study the bonds of silicon in its compounds, the heats of the following reactions in the gaseous state at very high temperatures were measured in a pump calorimeter (at constant volume), and using the First Principle of Thermodynamics, the enthalpy changes were calculated.

**13.1.** What thermodynamic magnitude can be determined directly in a pump calorimeter?**13.2.** Calculate the bond energies E (in kJ/mol) of Si-Cl, Si-C, Si-O and C-Cl, given the values $E(\text{O-H}) = 463 \text{ kJ/mol}$, $E(\text{Cl-Cl}) = 243 \text{ kJ/mol}$ and $E(\text{Cl-H}) = 431 \text{ kJ/mol}$.

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Silanes ($\text{Si}_n\text{H}_{2n+2}$) are the analogs of alkanes but derived from silicon. Silanes consist of a chain of silicon atoms bonded to hydrogen atoms. The atomization energy (ΔH_{atom}) is the energy needed to break all the bonds of 1 mole of compound, forming only atoms in the gaseous state. For silanes, ΔH_{atom} depends on the number of silicon atoms, as shown below.

Silane	Si_4H_{10}	Si_5H_{12}	Si_6H_{14}	Si_7H_{16}	Si_8H_{18}	Si_9H_{20}
ΔH_{atom} (kJ/mol)	3900	4700	6000	6500	7400	8200

13.3. Find a linear equation that relates ΔH_{atom} to the **n** number of silicon atoms in the formula of a silane ($\text{Si}_n\text{H}_{2n+2}$) and to the energies of the bonds Si-Si and Si-H.

13.4. Calculate $E(\text{Si-Si})$ and $E(\text{Si-H})$ by linear regression or graphically, **as accurately as possible**. Register your calculations and reasoning.

13.5. Represent the structures of silicon hydride ($n = 1$), disilane ($n = 2$), and trisilane ($n = 3$). Calculate your atomization energies.

Methane (CH_4) is stable in air at room temperature and requires an ignition source to initiate combustion. In contrast, its silane analog is pyrophoric, meaning it spontaneously combusts upon contact with air. The structure of the reaction products also differs; while CO_2 is molecular and has two C=O bonds, SiO_2 is atomic and has four Si-O bonds per structural unit.

Enlace	C-H	C=O	O-H	O=O	Si-H	Si-O
E (kJ/mol)	413	799	467	498	318	463

13.6. Estimate the enthalpies of combustion for both substances and say which reaction is more favored.

Ethane ($\text{CH}_3\text{-CH}_3$) is a very stable molecule, with the C-C bond notoriously inert toward water and dilute bases. However, its disilane analog reacts easily with water or aqueous potassium hydroxide, releasing hydrogen.

13.7. Explain the difference in the reactivity of the two substances.

13.8. Write the balanced chemical equation for the hydrolysis reaction of disilane with hydroxide.

13.9. Trimethylamine $\text{N}(\text{CH}_3)_3$ has a pyramidal geometry on the nitrogen atom. However, the tris(trimethylsilyl)amine $\text{N}(\text{Si}(\text{CH}_3)_3)_3$ is trigonal flat. Explain the origin of the different geometries.

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